

MUZHI, Russian Federation

WMO: 234260

Lat: 65.404N

Lon: 64.703E

Elev: 19

StdP: 101.10

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.3	-38.5	-43.5	0.1	-40.9	-40.8	0.1	-38.4	11.0	-12.6	9.5	-11.5	1.0	10	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	26.2	19.3	23.8	18.0	21.6	16.7	20.4	24.4	18.9	22.6	17.4	20.8	3.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	13.6	22.9	17.4	12.4	21.0	15.8	11.2	19.7	58.7	24.3	53.8	22.7	49.1	20.9	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.3	8.1	DB	-43.6	29.6	3.1	1.8	-45.8	30.8	-47.6	31.9	-49.4	32.9	-51.7	34.2	(4)
(5)			WB	-43.4	22.6	2.8	1.8	-45.4	23.9	-47.0	24.9	-48.6	25.9	-50.6	27.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.7	-21.8	-21.2	-12.2	-4.9	2.3	11.3	15.7	12.1	6.4	-1.4	-13.0	-18.8	(6)	
	DBStd	14.68	9.16	9.38	7.91	6.74	5.65	4.95	4.28	3.60	3.86	5.34	8.72	9.63	(7)	
	HDD10.0	5362	986	873	687	447	247	43	4	17	120	353	691	894	(8)	
	HDD18.3	8069	1244	1106	946	696	496	217	106	197	358	611	941	1153	(9)	
	CDD10.0	367	0	0	0	1	10	82	181	81	12	1	0	0	(10)	
	CDD18.3	33	0	0	0	0	0	6	24	3	0	0	0	0	(11)	
	CDH23.3	239	0	0	0	0	2	46	172	20	1	0	0	0	(12)	
	CDH26.7	43	0	0	0	0	0	7	34	3	0	0	0	0	(13)	
(14) Wind	WSAvg	3.2	2.6	2.7	3.2	3.7	3.9	3.9	3.5	3.3	3.2	3.2	2.9	2.6	(14)	
(15) Precipitation	PrecAvg	546	28	24	40	39	45	59	76	69	47	56	36	28	(15)	
	PrecMax	678	51	50	264	81	109	129	176	178	126	86	71	57	(16)	
	PrecMin	447	8	3	5	10	14	9	30	20	14	7	4	9	(17)	
	PrecStd	63	12	12	46	18	22	31	33	40	24	19	16	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	0.1	5.2	13.4	22.7	27.3	29.8	25.9	21.0	12.5	3.6	0.6	(19)	
		MCWB	-1.1	-1.4	1.8	8.1	14.0	18.5	20.9	18.5	15.6	9.5	1.8	-1.0	(20)	
	2%	DB	-3.9	-2.6	2.7	7.9	18.2	24.4	27.4	22.2	16.8	9.4	1.5	-1.6	(21)	
		MCWB	-5.0	-3.7	0.2	4.1	11.8	17.0	20.1	17.2	13.1	7.2	0.5	-2.6	(22)	
	5%	DB	-6.3	-5.0	1.1	5.7	14.3	21.7	25.2	20.2	14.1	7.2	0.1	-3.8	(23)	
		MCWB	-7.1	-5.8	-0.8	2.6	9.4	15.8	19.3	15.8	11.4	5.4	-0.8	-4.5	(24)	
	10%	DB	-8.7	-7.2	-0.8	3.9	10.7	19.5	23.0	18.3	12.1	5.1	-1.6	-6.0	(25)	
		MCWB	-9.3	-8.0	-2.4	1.6	7.0	14.8	18.3	14.9	9.8	3.7	-2.5	-6.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	-1.3	2.2	8.7	14.5	20.3	22.8	20.0	16.2	9.9	1.9	-0.7	(27)	
		MCDB	0.0	0.1	4.8	13.5	20.6	24.7	27.1	24.0	20.1	12.0	2.6	0.3	(28)	
	2%	WB	-4.8	-3.5	0.6	4.5	11.9	18.1	21.4	17.9	13.6	7.2	0.7	-2.6	(29)	
		MCDB	-4.0	-2.6	2.4	7.3	17.6	22.7	25.6	21.3	16.2	9.0	1.4	-1.7	(30)	
	5%	WB	-7.1	-5.8	-0.8	3.0	9.6	16.5	20.0	16.5	11.7	5.5	-0.8	-4.4	(31)	
		MCDB	-6.4	-5.1	1.0	5.1	14.0	21.1	24.0	19.3	13.7	7.0	0.0	-3.8	(32)	
	10%	WB	-9.3	-8.1	-2.4	1.8	7.2	15.0	18.8	15.2	10.2	3.9	-2.6	-6.6	(33)	
		MCDB	-8.7	-7.2	-0.8	3.8	10.5	18.9	22.5	17.8	11.8	5.0	-1.7	-6.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.2	9.3	8.6	7.6	7.6	8.3	7.8	6.5	5.2	7.1	7.9	(35)	
		MCDBR	8.9	8.4	8.9	9.1	13.1	11.5	11.4	10.8	9.8	6.8	6.6	8.6	(36)	
	5% WB	MCWBR	8.4	7.6	7.1	6.5	8.0	6.3	6.2	6.7	6.4	5.1	5.8	7.9	(37)	
		MCWBR	8.6	8.4	8.7	8.1	12.6	10.2	10.1	9.6	9.0	6.7	6.2	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.206	0.266	0.313	0.340	0.352	0.368	0.411	0.375	0.328	0.286	0.204	0.213	(40)	
		taud	1.893	2.048	2.051	2.091	2.293	2.435	2.298	2.376	2.474	2.280	1.931	1.902	(41)	
	Edn at Noon		257	592	732	814	849	845	786	782	749	584	249	44	(42)	
		Edh at Noon	22	61	98	121	110	99	110	92	67	48	20	7	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.10	0.72	2.28	4.23	5.37	5.25	5.27	3.45	1.88	0.73	0.18	0.03	(44)	
	RadStd		0.01	0.07	0.14	0.27	0.36	0.41	0.51	0.28	0.19	0.09	0.02	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+1.16	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page